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AI for Bharat Hackathon

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Team Name : **Bit-Storm**

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Problem Statement : **Design an AI solution that improves efficiency, understanding, or support within healthcare or life-sciences ecosystems**.

❖ Problem:

Today, critical medical information such as allergies, chronic conditions, medications, and blood group is scattered across hospitals, prescriptions, and personal documents. During emergencies, patients may be unconscious or unable to communicate, leaving doctors without instant access to verified medical history. Most existing health record systems depend on internet connectivity and hospital-specific logins, making them unreliable in urgent situations. They also fail to provide a quick, structured medical overview focused on emergencies. This fragmentation causes treatment delays, medication errors, and increased risk to patient safety.

❖ Brief about the Idea:

We are building Dr.One – a Smart Health Card system designed to provide instant, secure, and offline access to essential medical information.
At the core of the solution is a patient-owned digital health profile that securely stores:

- Medical records and prescriptions
- Allergies and chronic conditions
- Current medications

It also supports **real wearable device integration**, securely collecting contextual physiological data (heart rate, SpO_2, activity) with user consent.
An AI-powered assistant generates a clear medical summary to help healthcare professionals quickly understand a patient's condition without diagnosing or prescribing.

➢ The system works as follows:

Dr.One enables:

- Emergency QR access to critical medical data (works offline)
- AI-assisted medical summaries for faster decision-making
- Append-only medical records to ensure data integrity
- Privacy-first design, where users fully own and control their data

➢ The solution is designed to be:

- **Emergency-ready:** instant access without login
- **Secure & private:** read-only access for medical staff
- **AI-assisted:** summaries for clarity, not diagnosis
- **Scalable & future-ready:** supports integration with hospitals and national health systems

❖ Differ From Others

How Our Solution Clearly Stands Apart

Emergency-First & Offline by Design

- Instant access to critical medical data **without internet**
- QR-based emergency scan works even if hospital systems are down
- No dependency on hospital logins or platforms

Built for real emergencies, not ideal conditions.

Responsible AI That Assists, Not Replaces

- AI converts complex medical records into **clear, actionable summaries**
- Highlights **allergies**, conditions, medications & risk alerts
- **No diagnosis**, no prescription — strictly decision support

AI helps doctors think faster, not decide for them.

True Patient Ownership & Privacy

- Patients **fully own** and **control** their health data
- Doctors get **read-only emergency** access only
- No tracking, no data resale, no hidden sharing

Trust is built into the system, not added later.

Designed for emergencies. Powered by ethical AI. Controlled by patients.

❖ Innovation and Uniqueness

CORE SAFETY FEATURES

Offline Emergency QR
Instant access without internet In critical emergencies
- Login-free QR retrieval in seconds

AI-Powered Medical Summary
- AI generates a clear review of allergies, meds, conditions
- Assist advisory only – no diagnosis

Emergency-First Design
- Built for real-world emergency speed & clarity.
- Minimizes data overload,

Privacy-First Patient-Owned Data
- Read-only access with strict privacy controls
- No data staring or hospital lock-in

Emergency-First Design
- Built for real-world emergency speed & clarity
- Minimizes data overload, priorities eter

Contextual Prescription Safety
- Detects when new prescriptions may conflict with current health state
- Advisory alerts based on medical

INTELLIGENT EXTENSIONS

Wearable Device Integra-
- Secure Bluetooth link to real wearable devices
- Heart rate, SpO_2, activity & sleep data

Minimum Safe Dataset (MSD)
- Default emergency view with life-critical information only

Verified Doctor Finder
Location-based provider
- Uses verified directories, no AI ranking
- Navigation support only

Emergency-Ready. Context-Aware. Device-Integrated. Human-Controlled.

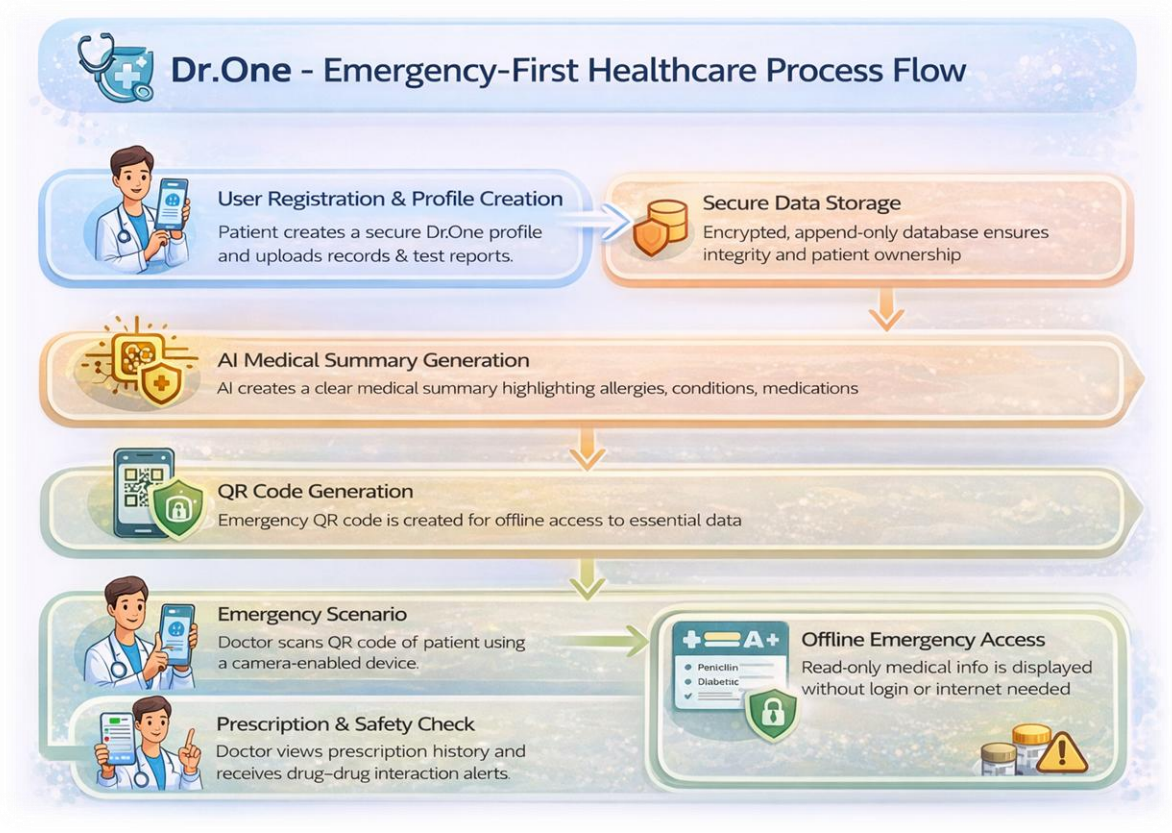
❖ How it Addresses the Problems

- ❑ Enables instant access to verified medical data during emergencies.
- ❑ Reduces medication errors using prescription history and interaction alerts.
- ❑ Eliminates dependency on internet and hospital-specific systems.

List of features offered by the solution:

1. **Digital Health Card (Dr.One ID)** with patient-owned medical profile
2. **Offline Emergency QR Access** for instant, login-free record viewing
3. **Secure, Append-Only Medical Records** with encryption and integrity
4. **AI-Generated Medical Summaries** for quick understanding (no diagnosis)
5. **Prescription History Management** with date & medication filters
6. **Drug–Drug Interaction Alerts** to prevent medication conflicts
7. **Vaccination & Test Report Storage** (COVID-19 and other diagnostics)
8. **Scalable, Privacy-First Platform** with future wearable/device integration

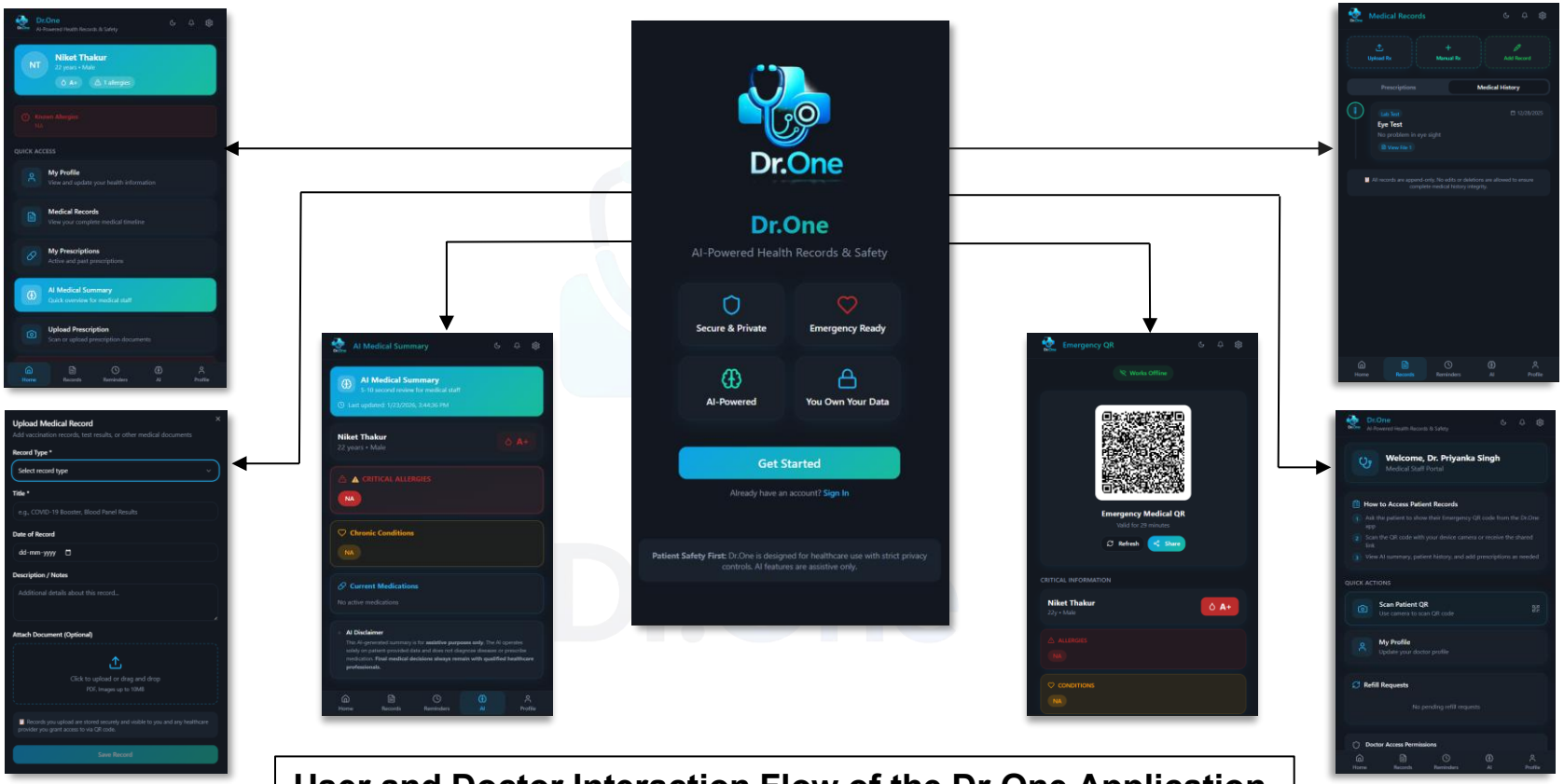
Process flow diagram or Use-case diagram:



Why This Process Matters ?

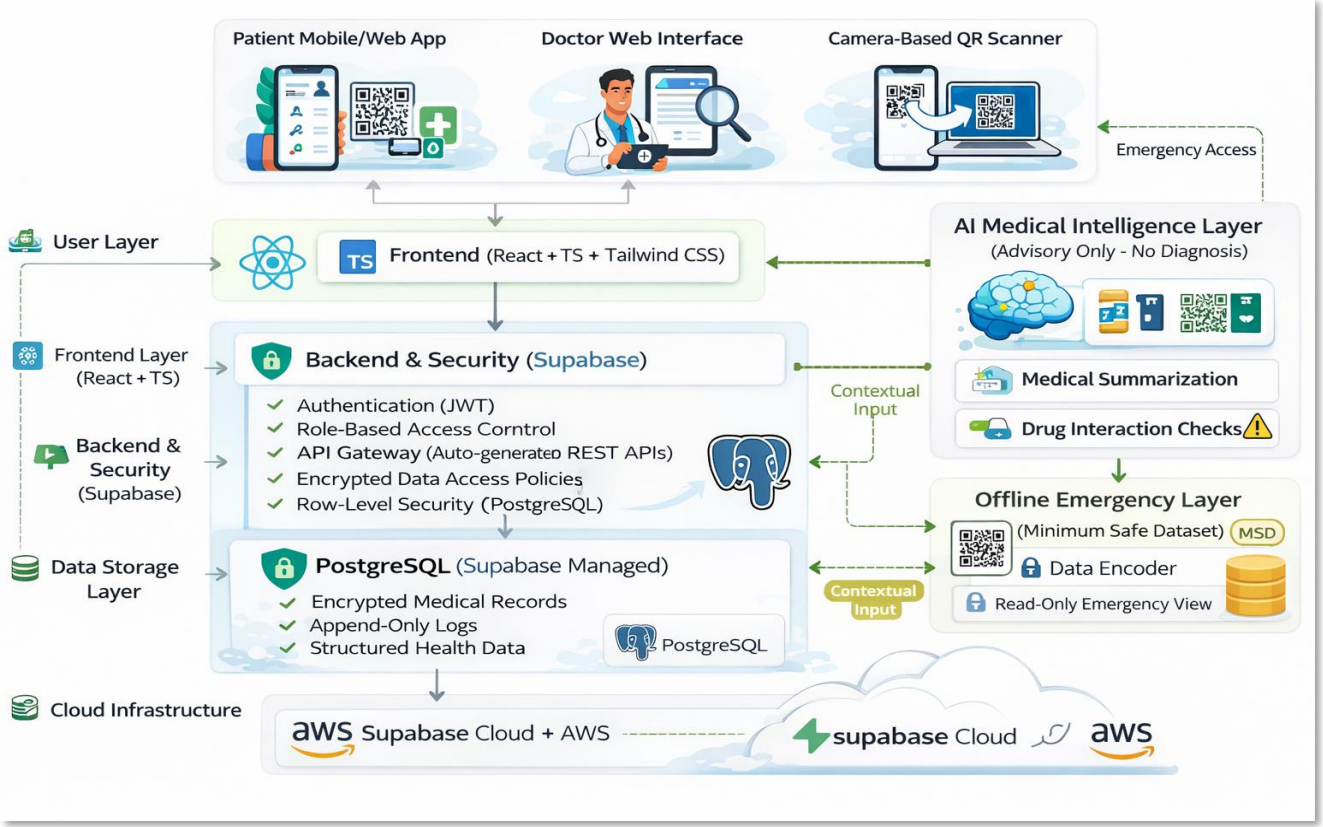
i. Eliminates delays caused by missing medical records.
ii. Works even when patients are unconscious or unable to communicate.
iii. Prevents medication errors using AI-assisted safety checks.
iv. Removes dependency on hospital-specific systems.
v. Ensures patient control and data privacy at every step.

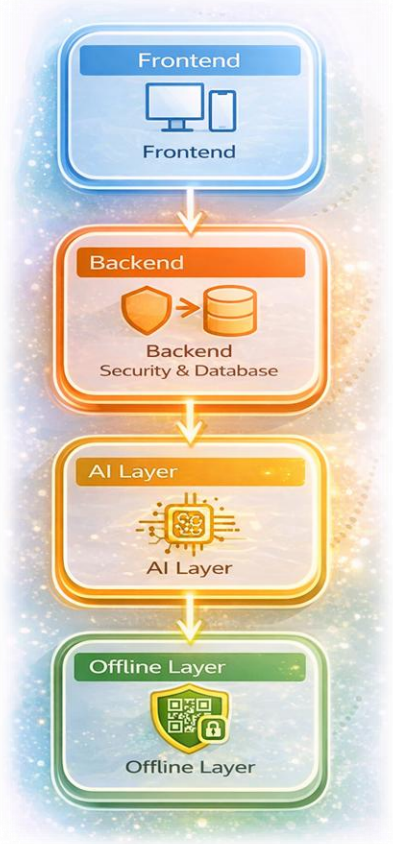
Wireframes/Mock diagrams of the proposed solution:



User and Doctor Interaction Flow of the Dr.One Application

Architecture diagram of the proposed solution:





Technologies to be used in the solution:

FRONTEND:

React | TS JavaScript | Tailwind CSS

BACKEND:

supabase

(Authentication, API, Storage)

DATABASE:

PostgreSQL

(PL/pgSQL Functions)

CLOUD & DEVOPS:

supabase Cloud

aws Cloud

PostgreSQL | docker

1. Data Collection & Storage

Collect medical records, prescriptions, and reports via secure user uploads.

Store structured data in Supabase-managed PostgreSQL with encrypted storage and append-only integrity to preserve medical history.

2. Backend & Security Layer

Leverage Supabase backend services for authentication, API management, and secure data access.

Implement role-based access control and enforce patient-owned data governance with encrypted communication.

3. AI Medical Summary Layer

Use transformer-based models to generate assistive medical summaries.

Extract key entities such as allergies, conditions, and medications (advisory only — no diagnosis or prescription).

4. Offline Emergency QR Layer

Generate QR-encoded critical medical data for offline access using the Minimum Safe Dataset (MSD).

Enable secure, read-only emergency view without login for authorized medical personnel.

5. Frontend & User Experience

Develop responsive mobile/web application using React with TypeScript and Tailwind CSS.

Design emergency-first UI focused on speed, clarity, safety, and seamless wearable integration.

Estimated implementation cost:

The solution is developed using **open-source technologies** and student-led development, keeping initial costs minimal.
Cloud infrastructure is supported using **AWS free tier / hackathon credits**, sufficient for prototype deployment.
AI-based medical summarization uses **lightweight transformer models**, avoiding heavy compute expenses.
QR-based offline emergency access eliminates the need for any specialized hardware or devices.

Estimated Prototype Cost:
₹5,000 – ₹10,000
Cost Effectiveness Highlights:

- No wearables or IoT devices required in the initial phase
- Low cloud and storage usage
- Scalable architecture suitable for large user bases at low incremental cost

30% — Open-Source & Student-Led Development

25% — AWS Free Tier & Cloud Credits

25% — Lightweight AI Models

20% — Offline QR & Emergency Access

Low cost, no hardware, open-source driven – designed for mass adoption.

Requirements for the hackathon:

1. The solution directly aligns with the *AI for Healthcare & Life Sciences* theme.
2. AI is used responsibly to summarize medical information, not to diagnose or prescribe.
3. It helps doctors and emergency staff understand patient history faster in critical moments.
4. Works even without internet, making it useful in rural and remote areas.
5. Patients fully own and control their medical data, ensuring privacy and trust.
6. The system is scalable and can integrate with hospitals and national health platforms.

Project Reference:

1. **World Health Organization (WHO)**
   Patient Safety and Emergency Care Systems
   https://www.who.int/news-room/fact-sheets/detail/patient-safety
2. **National Health Authority (India)**
   Ayushman Bharat Digital Mission (ABDM) – Digital Health Records & Privacy
   https://www.digitalindia.gov.in/initiative/ayushman-bharat-digital-mission/
3. **OpenMRS Community**
   Open-source Medical Record System for Low-Resource Settings
   https://openmrs.org
4. **GDPR & Data Privacy Principles**
   Patient-controlled data & consent-based access
   https://gdpr.eu
5. **Hugging Face & Transformer Models**
   Lightweight NLP models for summarization
   https://huggingface.co/docs/transformers
6. **AWS Security Best Practices**
   Cloud Security Documentation
   https://docs.aws.amazon.com/security
7. **HL7 FHIR Standard (Healthcare Interoperability)**
   Clinical data interoperability standard
   https://www.hl7.org/fhir/

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Thank You